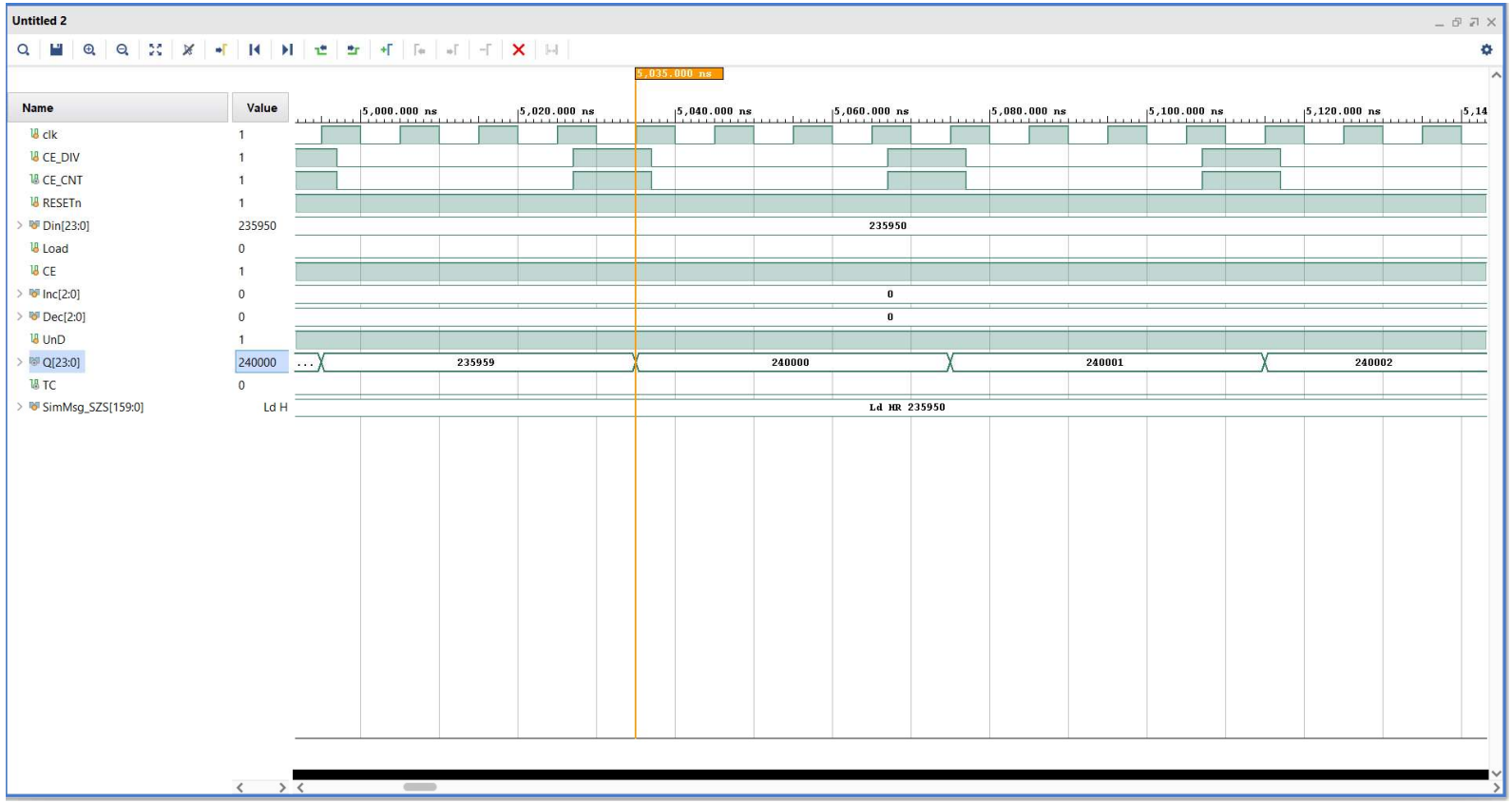
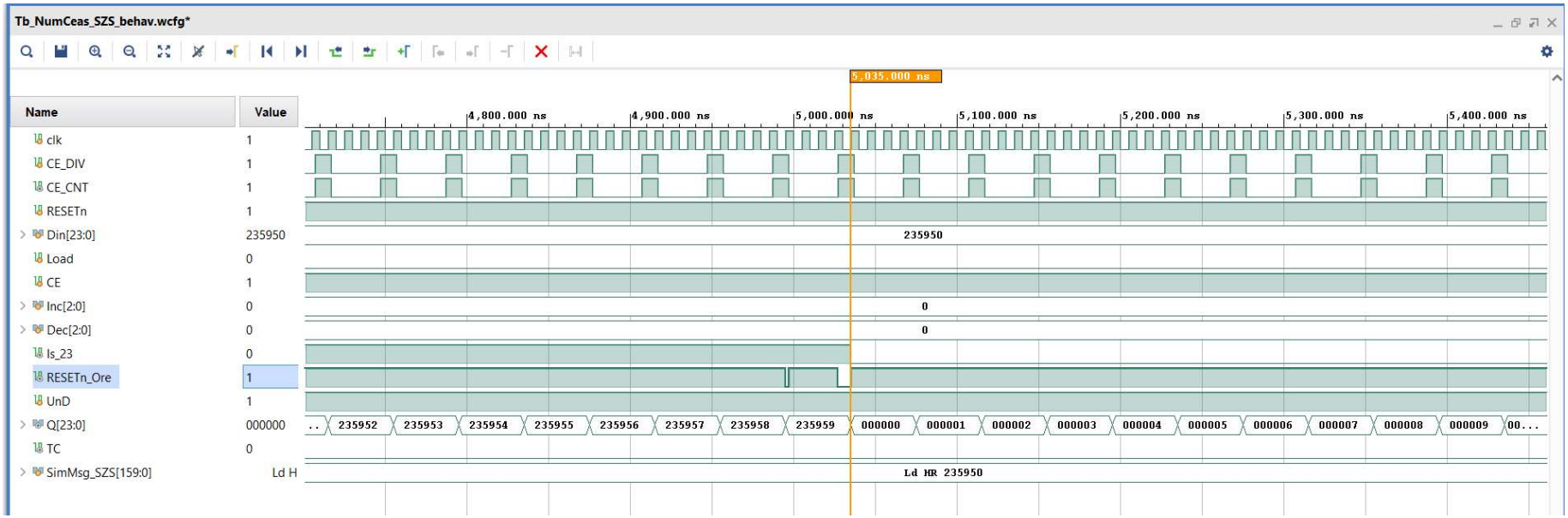
Digital clock

Covers hour-counter rollover, seconds adjustment, increment/decrement behavior and testbench waveforms. It also records debugging of an hour-counter reset problem, which can demonstrate your problem-solving process.

The hour counter does not reset

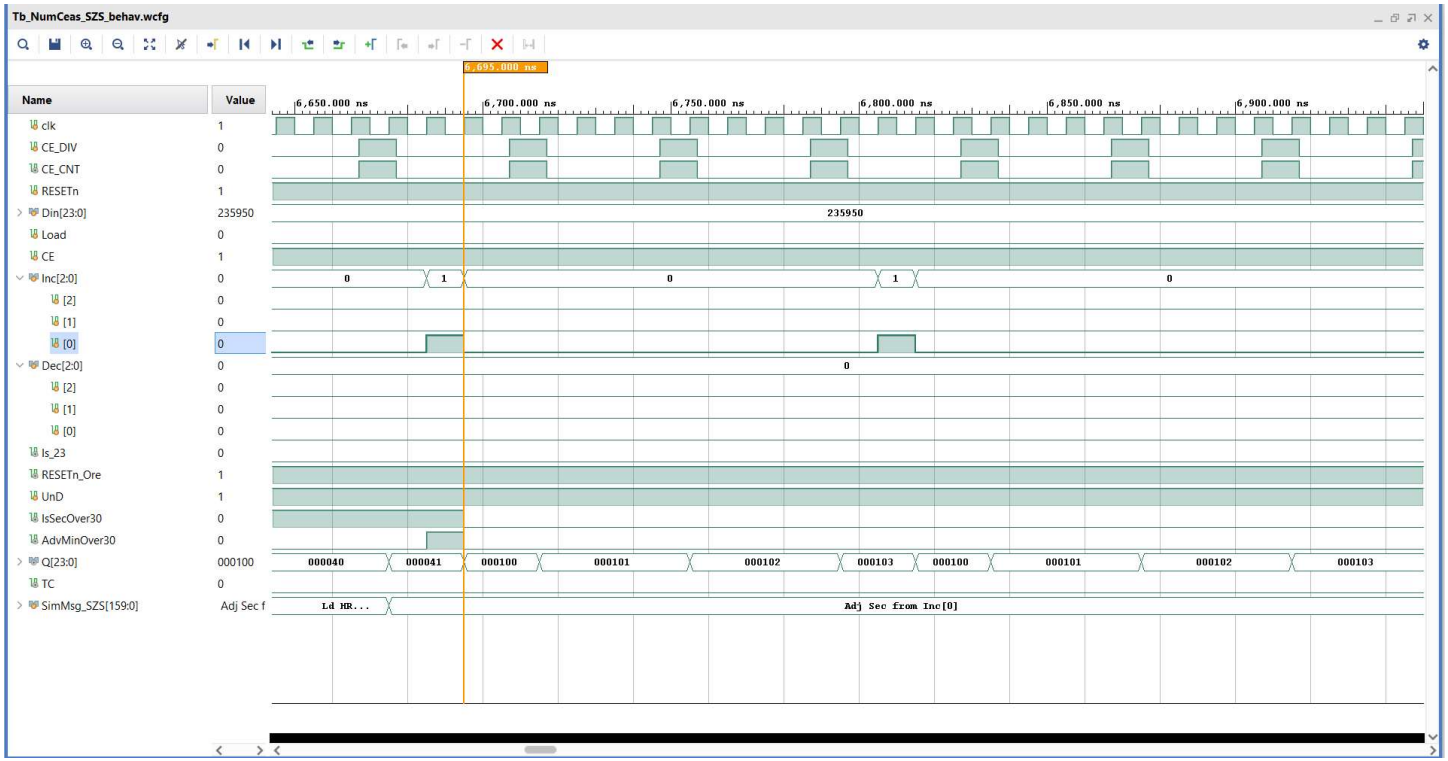


The hour counter resets




```
46  //wire [5:0] RSTn_Int_SZS = {6{RESETn}};
47  wire [5:0] RSTn_Int_SZS;
48
49  wire Is_23, RESETn_Ore;
50  assign Is_23 = (Q_SZS[23:16] == 8'h23)? 1'b1:1'b0;
51  assign RESETn_Ore = ( ~(Is_23 & CE_Int_SZS[4]) ) & RESETn;
52
53  assign RSTn_Int_SZS[3:0] = {4{RESETn}};
54  assign RSTn_Int_SZS[5:4] = {2{RESETn_Ore}};
55
```

Demonstrating that the seconds adjustment works correctly on Inc[0]



Code

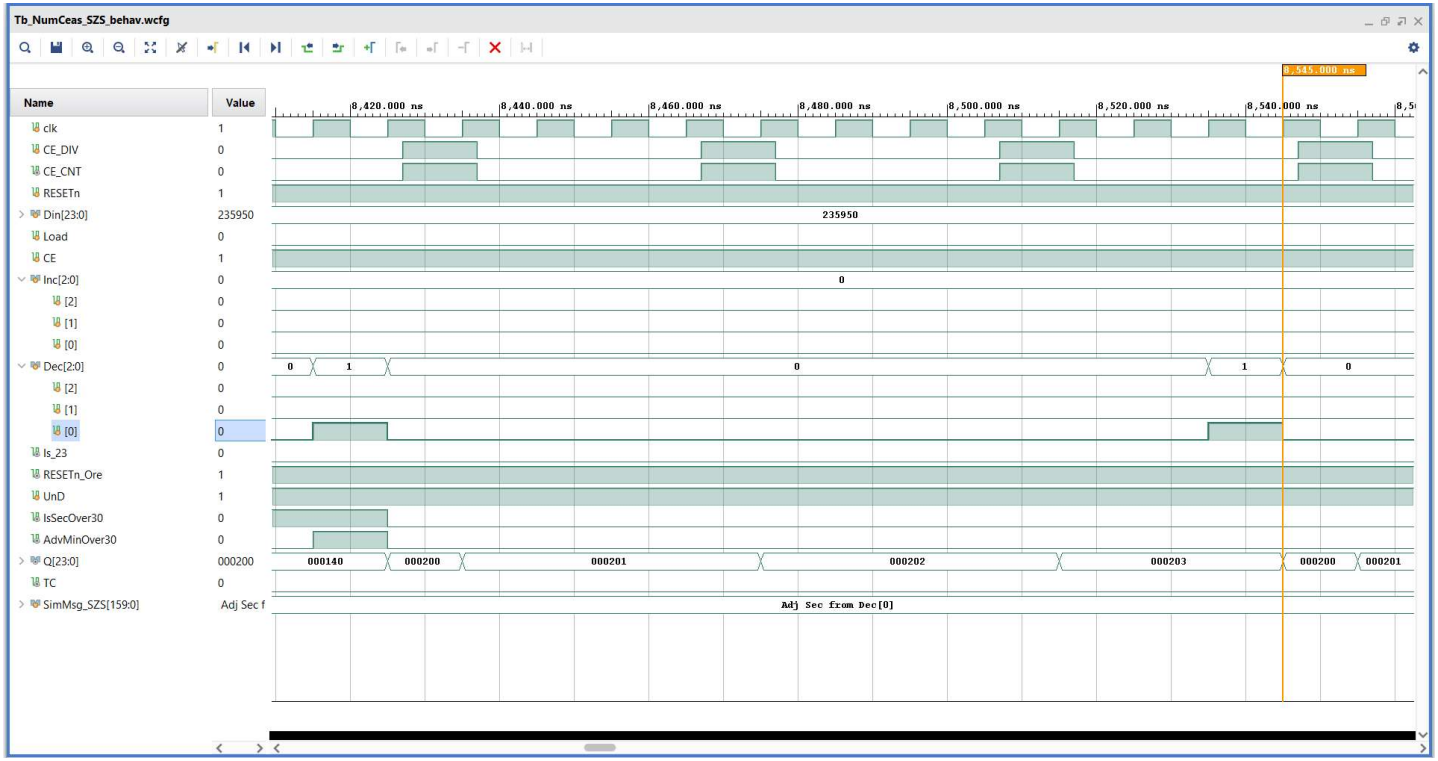
```
53  assign RSTn_Int_SZS[3:2] = {2{RESETn}};
54  assign RSTn_Int_SZS[5:4] = {2{RESETn_Ore}};
55
56  //Ajustare Secunde = Reset Secunde
57  assign RSTn_Int_SZS[0] = RESETn &
58      ( ~(Inc_SZS[0] | Dec_SZS[0]) );
59  assign RSTn_Int_SZS[1] = RSTn_Int_SZS[0];
60
61  //Reglaj secunde, avans minute la >30
62  wire IsSecOver30;
63  assign IsSecOver30 = (Q_SZS[7:0] >= 8'h30) ? 1'b1:1'b0;
64  wire AdvMinOver30 = (Inc_SZS[0] | Inc_SZS[0])& IsSecOver30;
65
```

```
83  //Minute
84  assign CE_Int_SZS[2] = (CE_Int_SZS[1] & TC_Int_SZS[1]) |
85                         AdvMinOver30 ;
```

TB_NumCeas

```
124 @(Q[7:0] == 8'h40);
125 SimMsg_SZS = "Adj Sec from Inc[0]";
126 repeat (2)begin
127     @ (posedge clk);
128     Inc[0] <= 1;
129     @ (posedge clk);
130     Inc[0] <= 0;
131     repeat (10) @ (posedge clk);
132 end
133
134 @(Q[7:0] == 8'h40);
135 @ (posedge clk);
136 SimMsg_SZS = "Adj Sec from Dec[0]";
137 repeat (2)begin
138     @ (posedge clk);
139     Dec[0] <= 1;
140     @ (posedge clk);
141     Dec[0] <= 0;
142     repeat (10) @ (posedge clk);
143 end
```

Demonstrating that the seconds adjustment works correctly on Dec[0]



Code

```
61  //Reglaj secunde, avans minute la >30
62  wire IsSecOver30;
63  assign IsSecOver30 = (Q_SZS[7:0] >= 8'h30) ? 1'b1:1'b0;
64  wire AdvMinOver30 = (Inc_SZS[0] | Dec_SZS[0])& IsSecOver30;
65
```

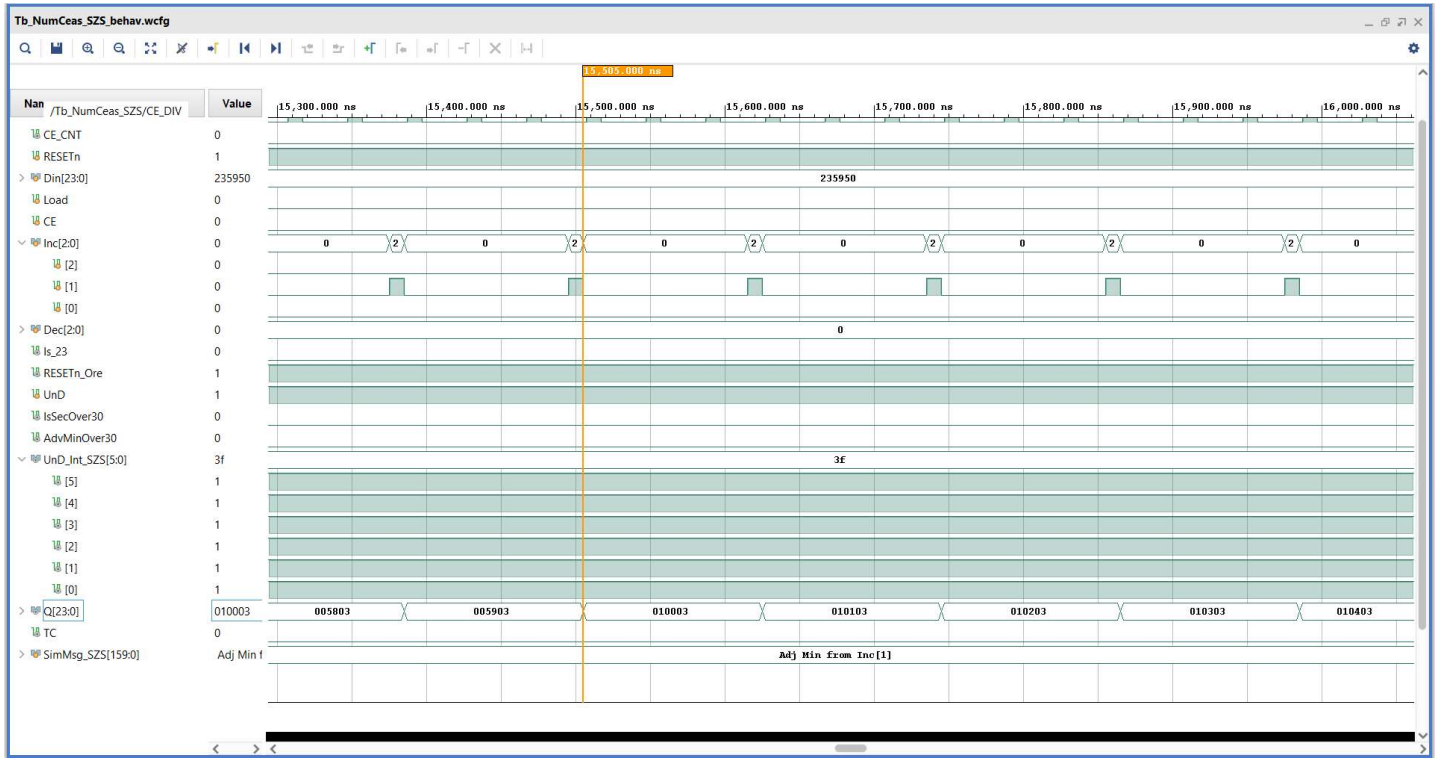
The logic of adjustment ports

```
66  //Logica UnD pentru ajustarea minutelor
67  assign UnD_Int_SZS[2] = (Inc_SZS[1])? 1'b1 :
68                          (Dec_SZS[1])? 1'b0 : UnD_SZS;
69  assign UnD_Int_SZS[3] = UnD_Int_SZS[2];
70
71  //UnD Secunde
72  assign UnD_Int_SZS[1:0] = {2{UnD_SZS}};
73  //UnD Ore
74  assign UnD_Int_SZS[5:4] = {2{UnD_SZS}};
75
83  //Minute
84  assign CE_Int_SZS[2] = (CE_Int_SZS[1] & TC_Int_SZS[1]) |
85                         AdvMinOver30 | Inc_SZS[1] | Dec_SZS[1];
```

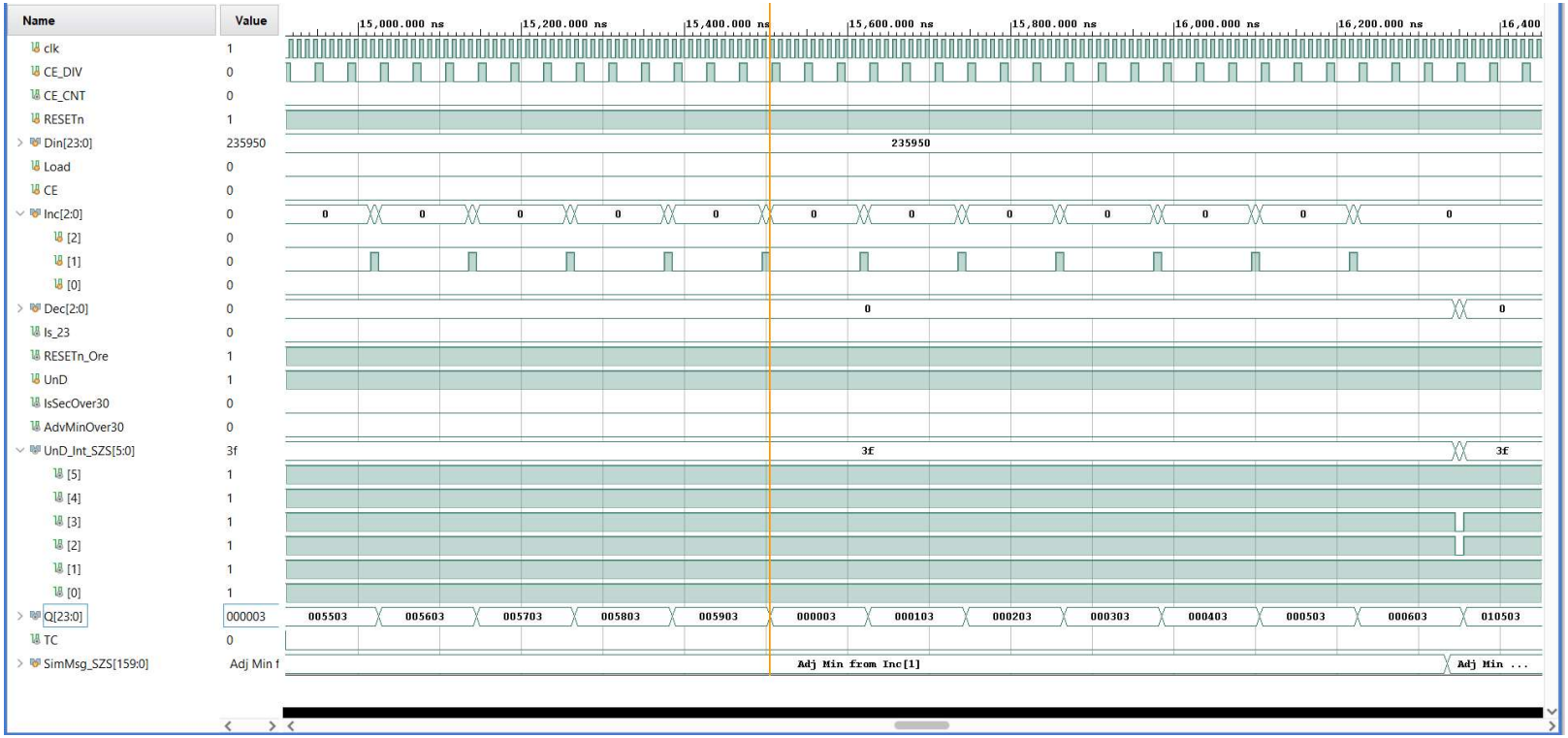
TB_NumCeas

```
146 //oprim ceasul sa putem urmari doar ajustarile
147 CE<=0;
148
149 SimMsg_SZS = "Adj Min from Inc[1]";
150 repeat (64)begin
151     @ (posedge clk);
152     Inc[1] <= 1;
153     @ (posedge clk);
154     Inc[1] <= 0;
155     repeat (10) @ (posedge clk);
156 end
157
158 @ (posedge clk);
159 SimMsg_SZS = "Adj Min from Dec[1]";
160 repeat (64)begin
161     @ (posedge clk);
162     Dec[1] <= 1;
163     @ (posedge clk);
164     Dec[1] <= 0;
165     repeat (10) @ (posedge clk);
166 end
```

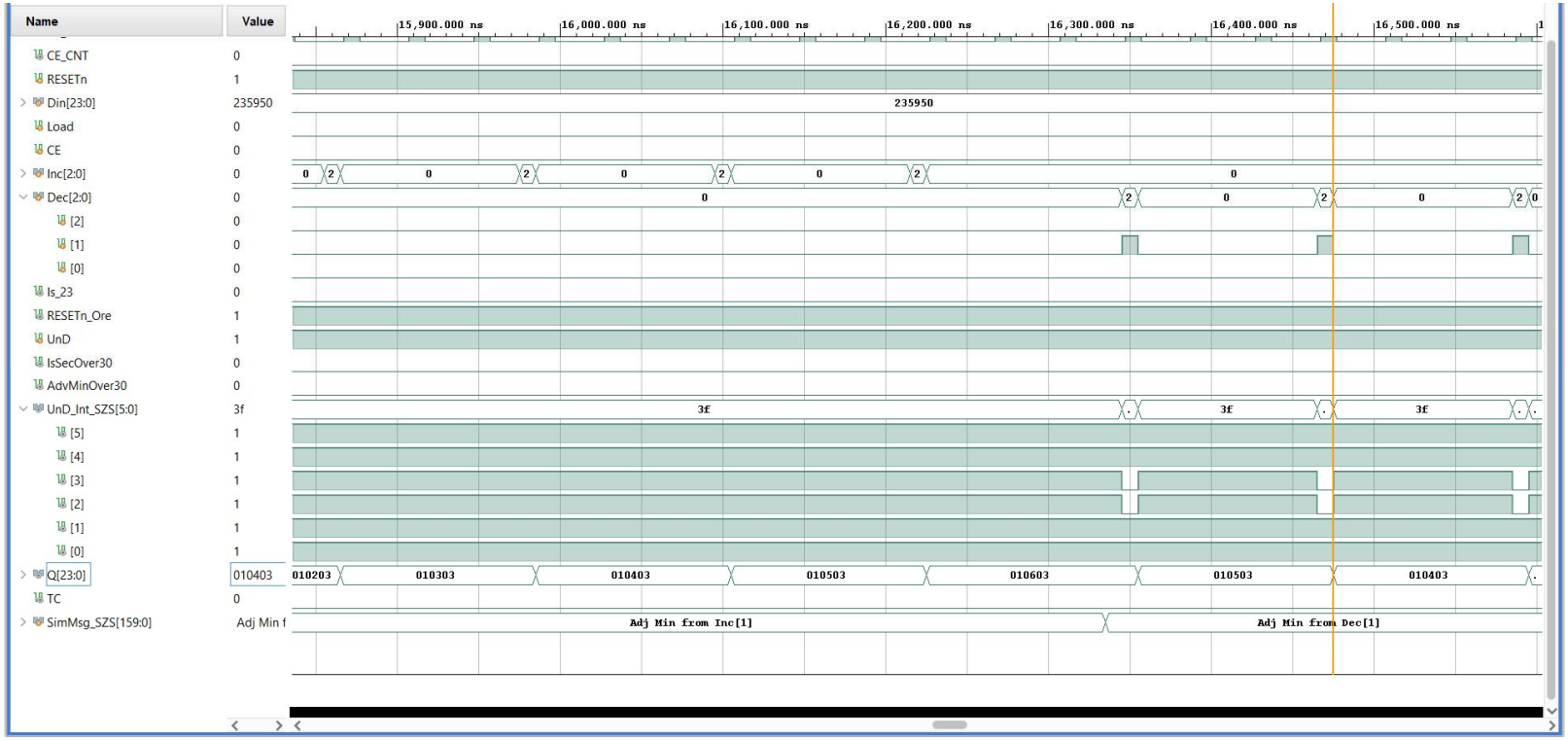
Increment – by mistake



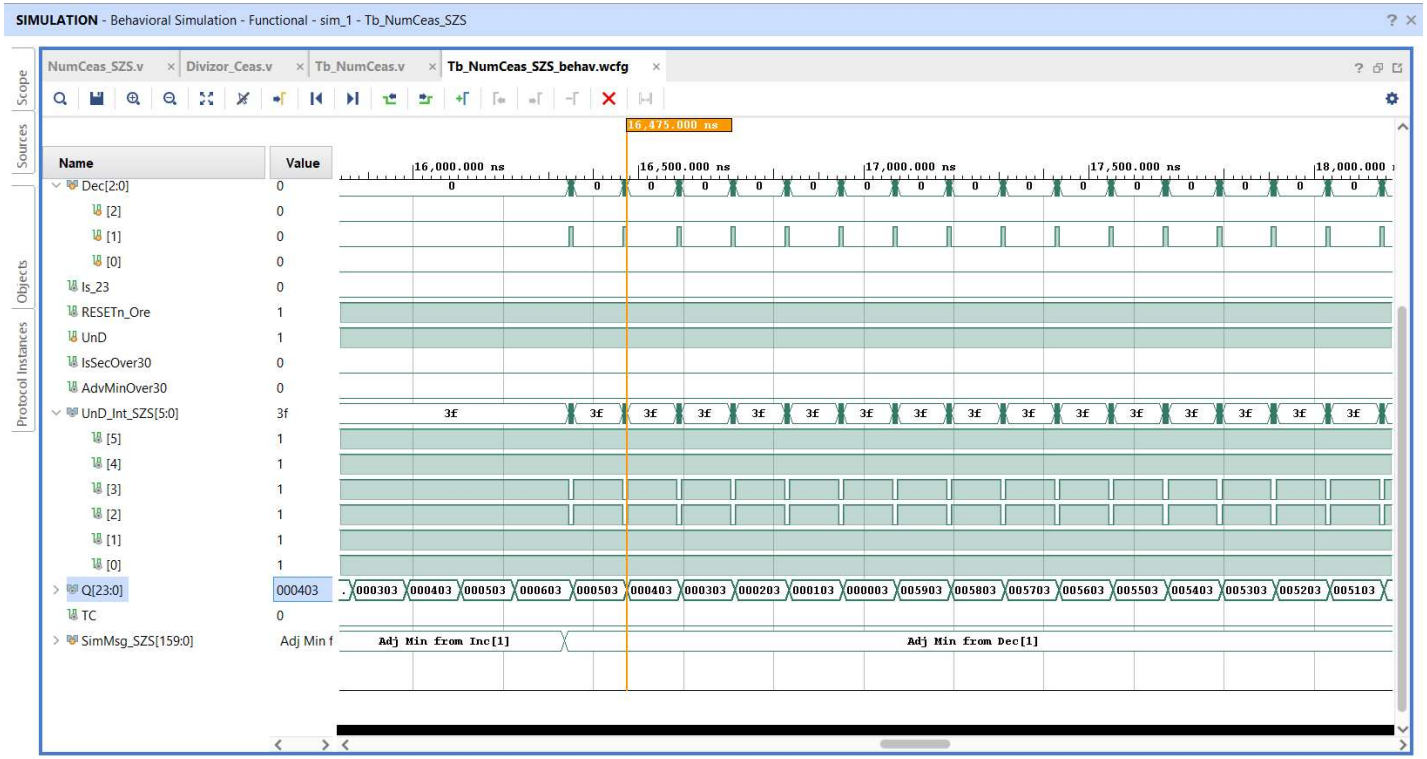
Increment – without error



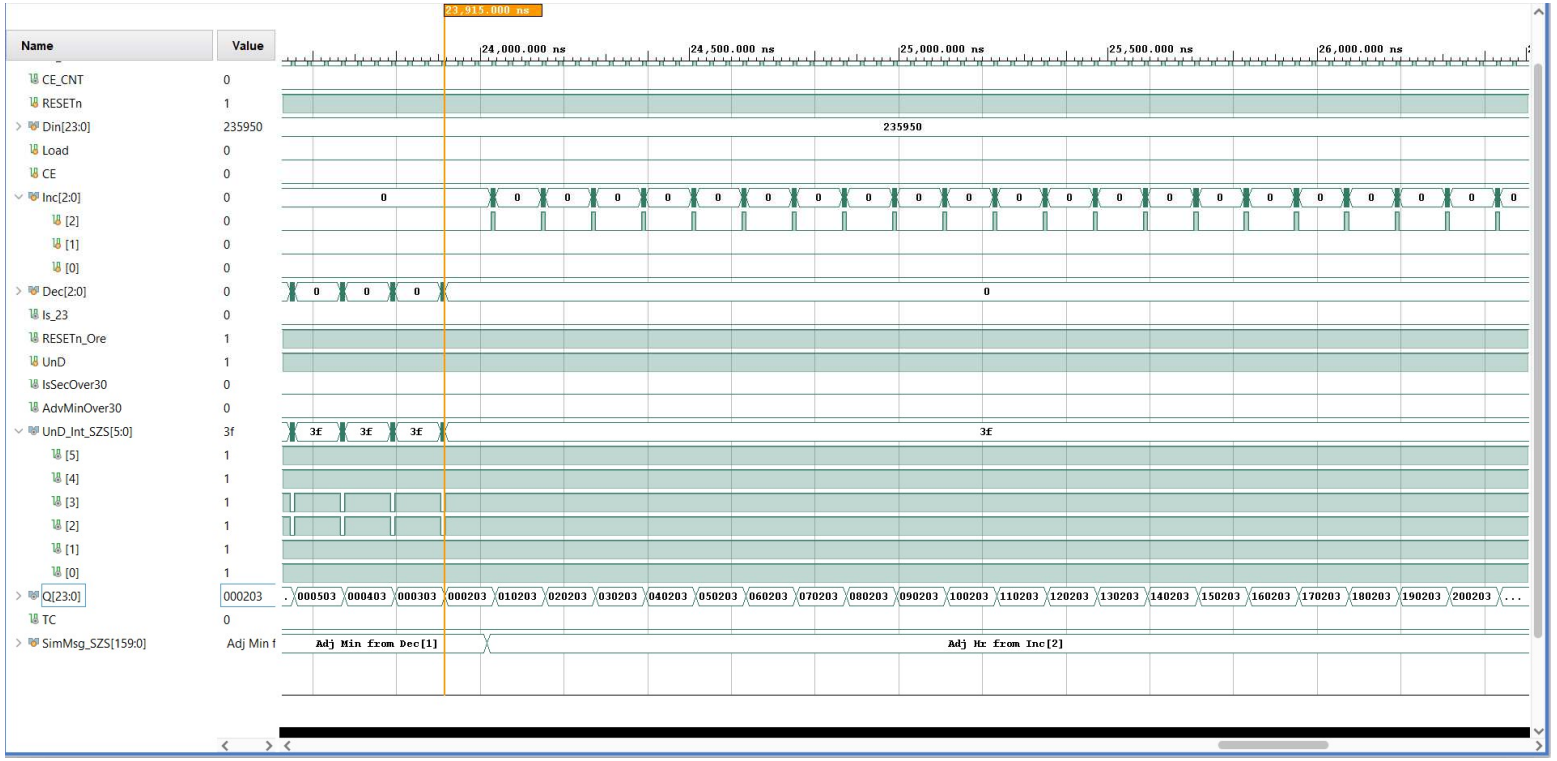
Decrement


Name
Value
15,900.000 ns
16,000.000 ns
16,100.000 ns
16,200.000 ns
16,300.000 ns
16,400.000 ns
16,500.000 ns
CE_CNT
0
RESETn
1
Din[23:0]
235950
Load
0
CE
0
Inc[2:0]
0
Dec[2:0]
0
[2]
0
[1]
0
[0]
0
Is_23
0
RESETn_Ore
1
UnD
1
IsSecOver30
0
AdvMinOver30
0
UnD_Int_SZS[5:0]
3f
[5]
1
[4]
1
[3]
1
[2]
1
[1]
1
[0]
1
Q[23:0]
010403
010203
010303
010403
010503
010603
010503
010403
TC
0
SimMsg_SZS[159:0]
Adj Min f
Adj Min from Inc[1]
Adj Min from Dec[1]

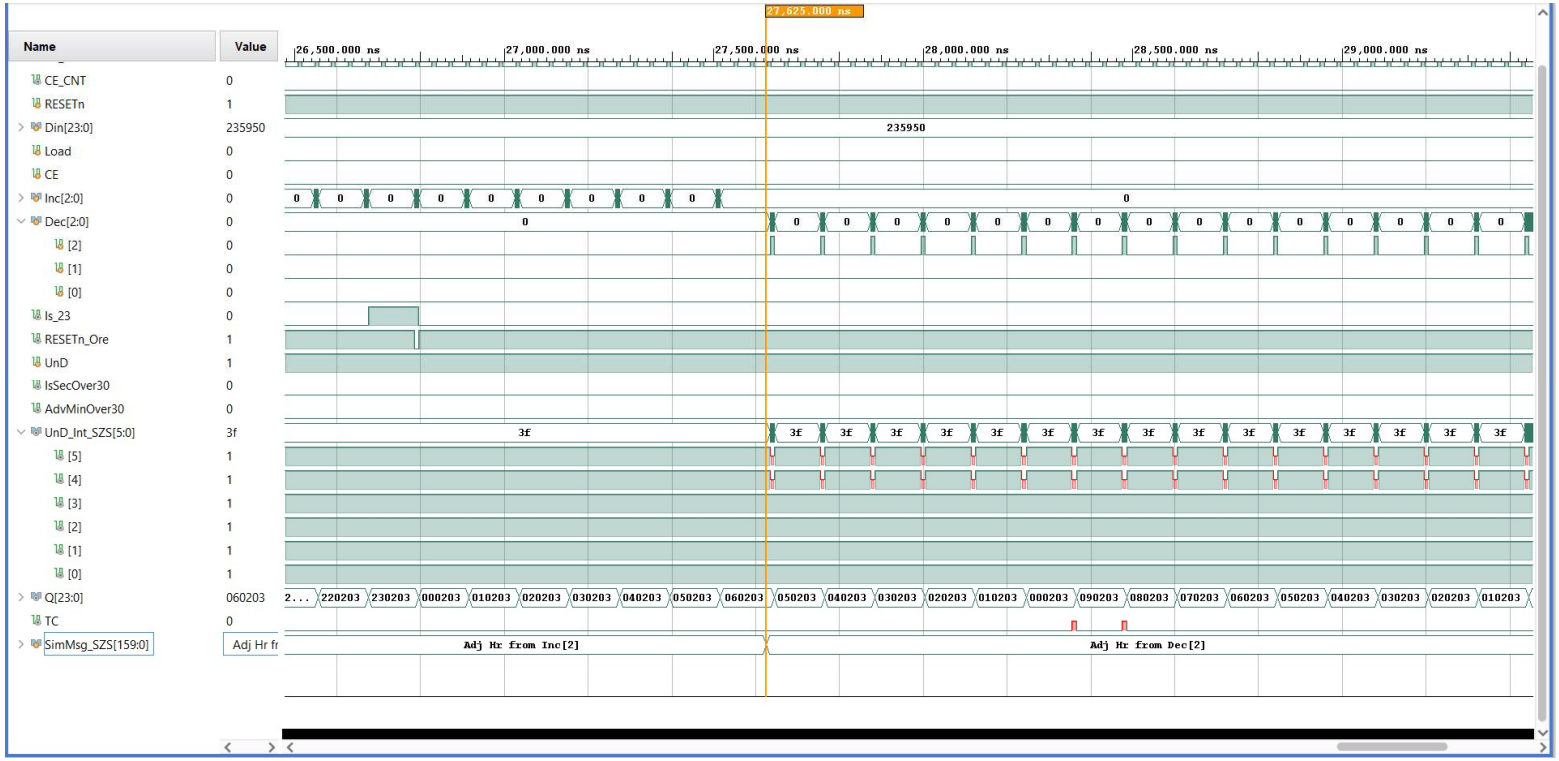
Decrement – without mistake



Time adjustment – increment



Time adjustment – decrement



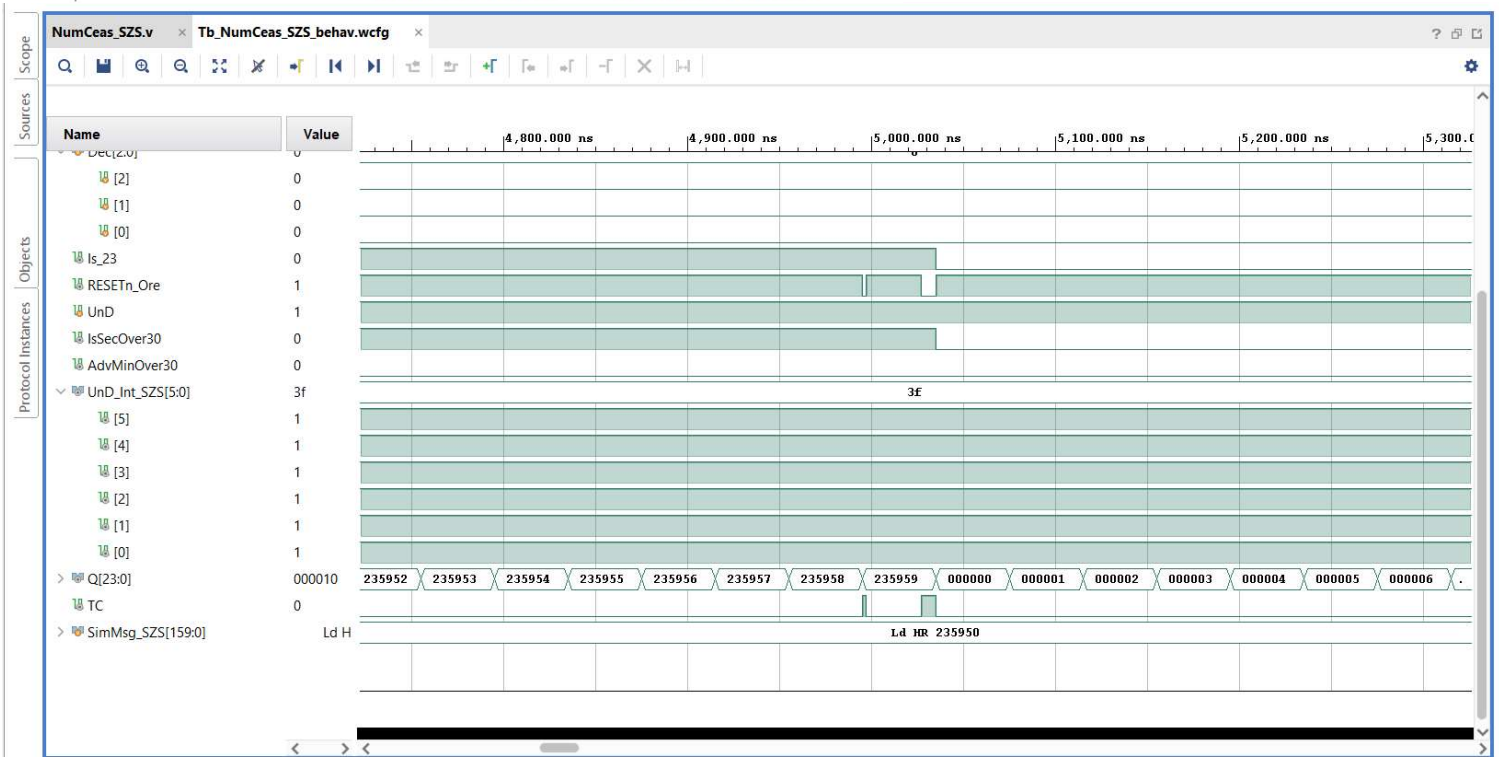
TC

```
100     //TC iesire
101     assign TC_SZS =   CE_Int_SZS[5] & TC_Int_SZS[5]
102                     & (~Inc_SZS[2]) & (~Dec_SZS[2]);
103
```

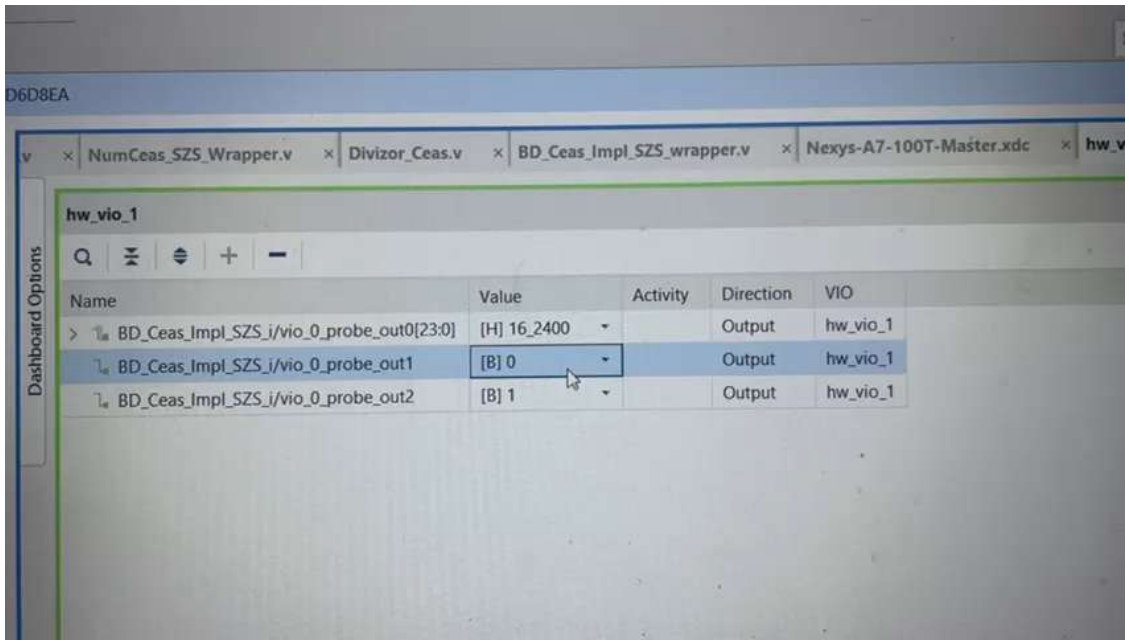


TC

```
100  //TC iesire
101  assign TC_SZS =    (Is_23 & CE_Int_SZS[4]) &
102                  & (~Inc_SZS[2]) & (~Dec_SZS[2]);
```



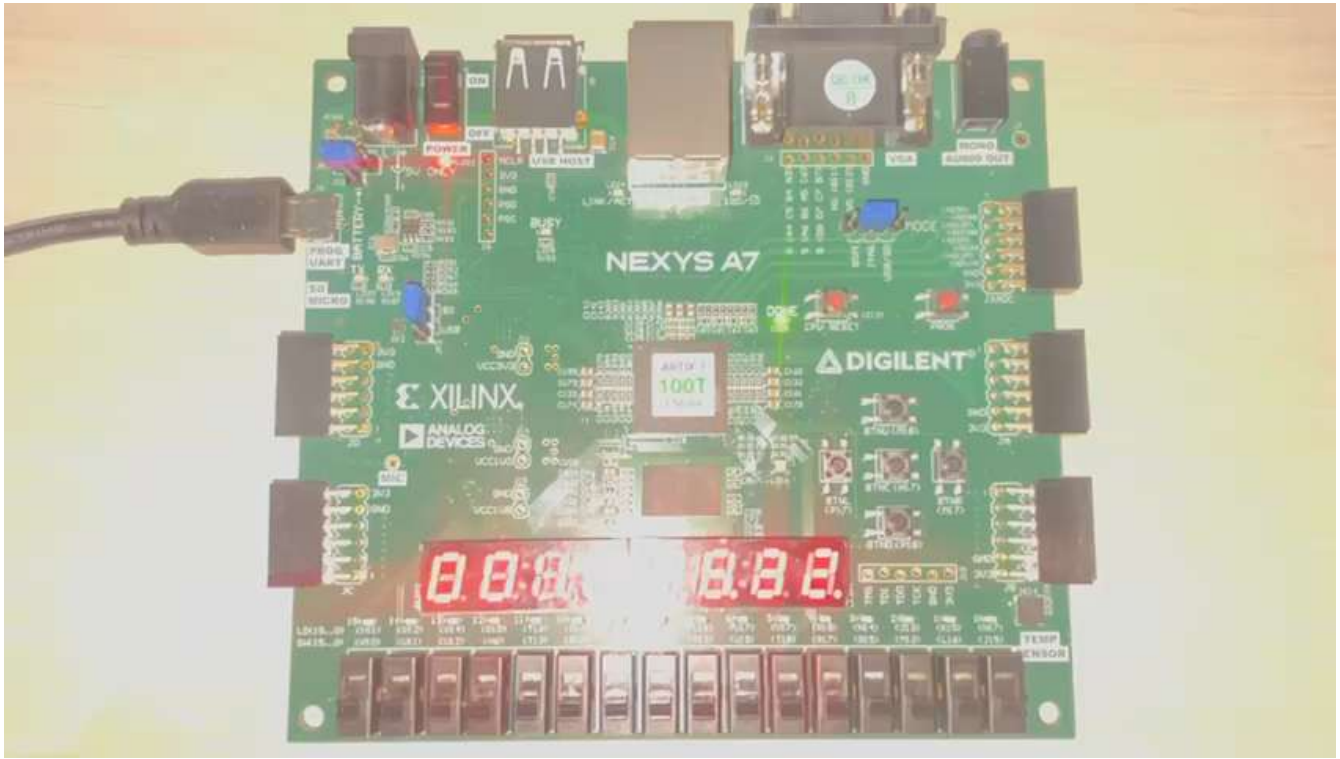
Setting the time to the current time using VIO (the DIN_SZS and Load signals)



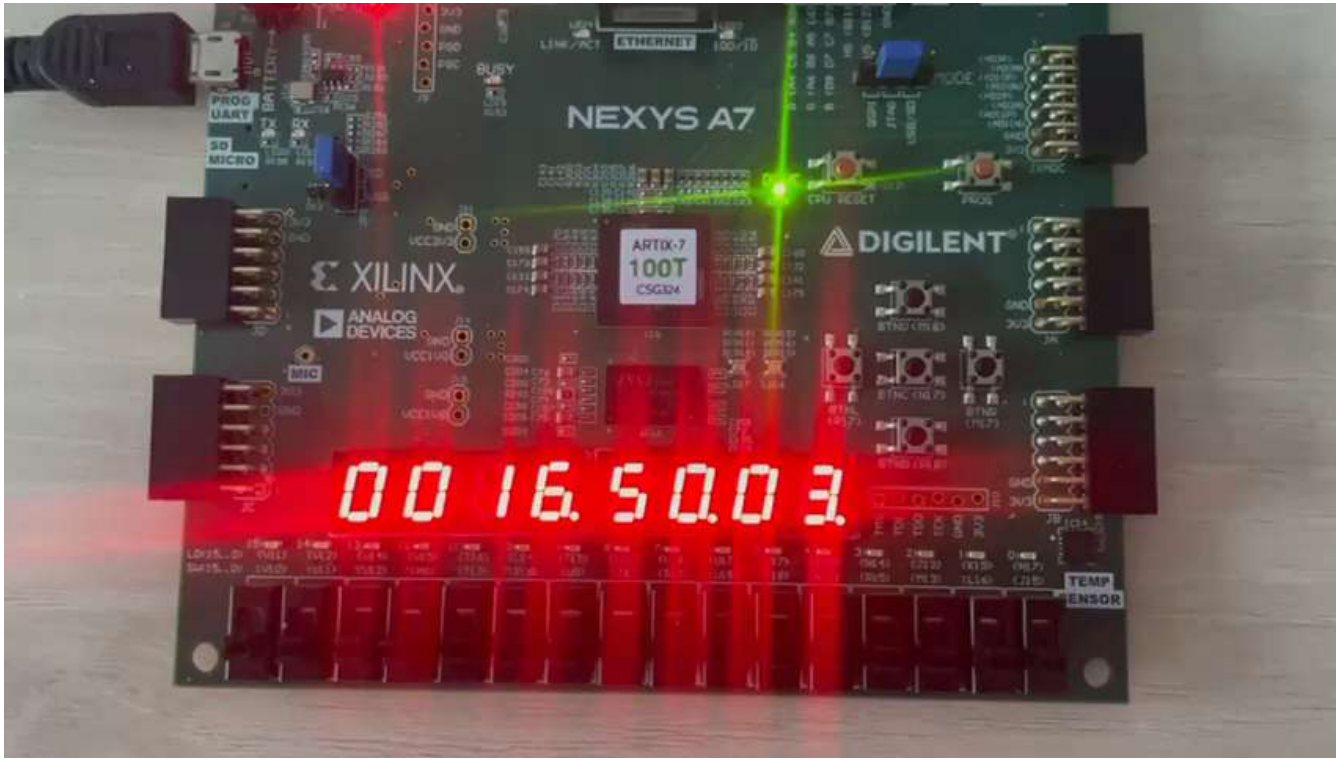
Adjusting (resetting) the seconds (from both SW[1] and SW[0])



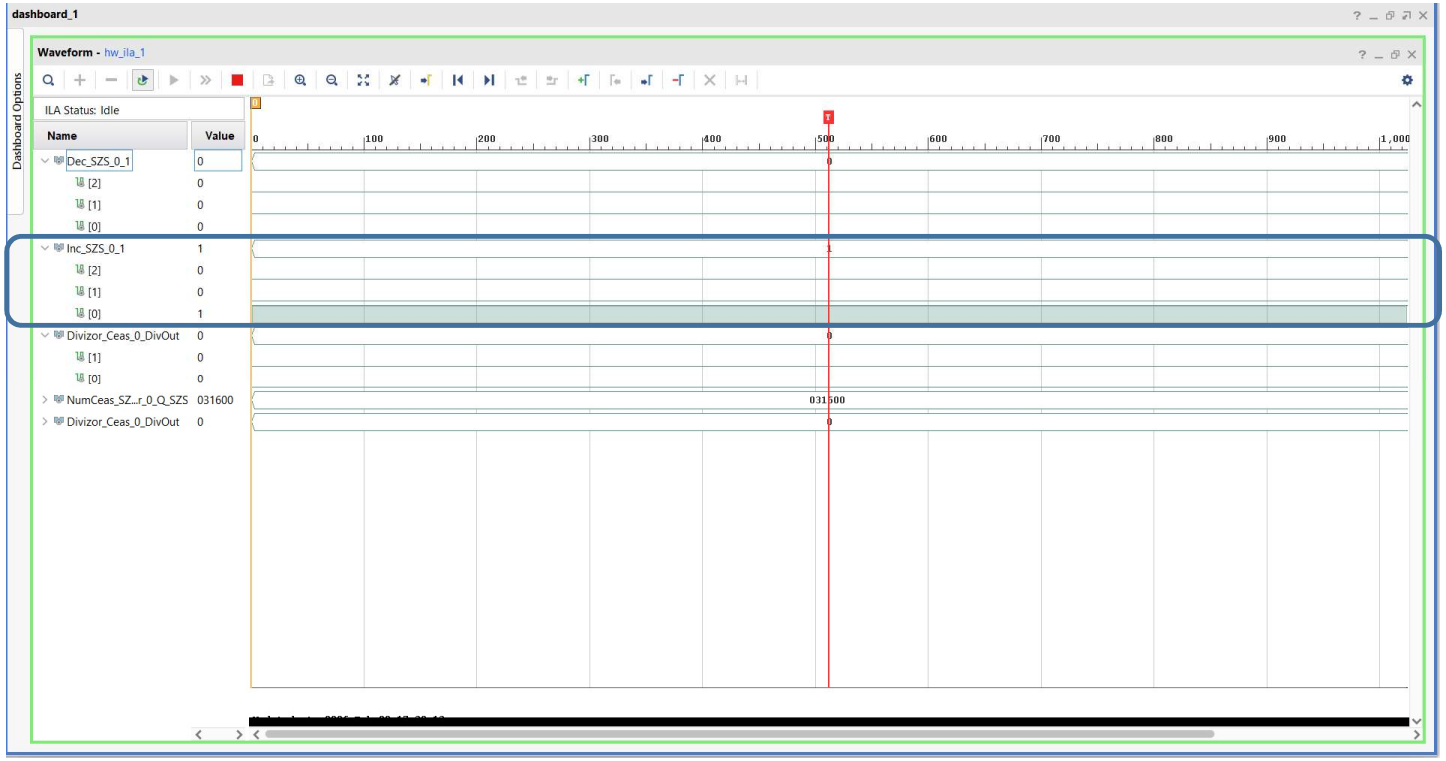
Adjusting the minutes (from BTNU and BTND)



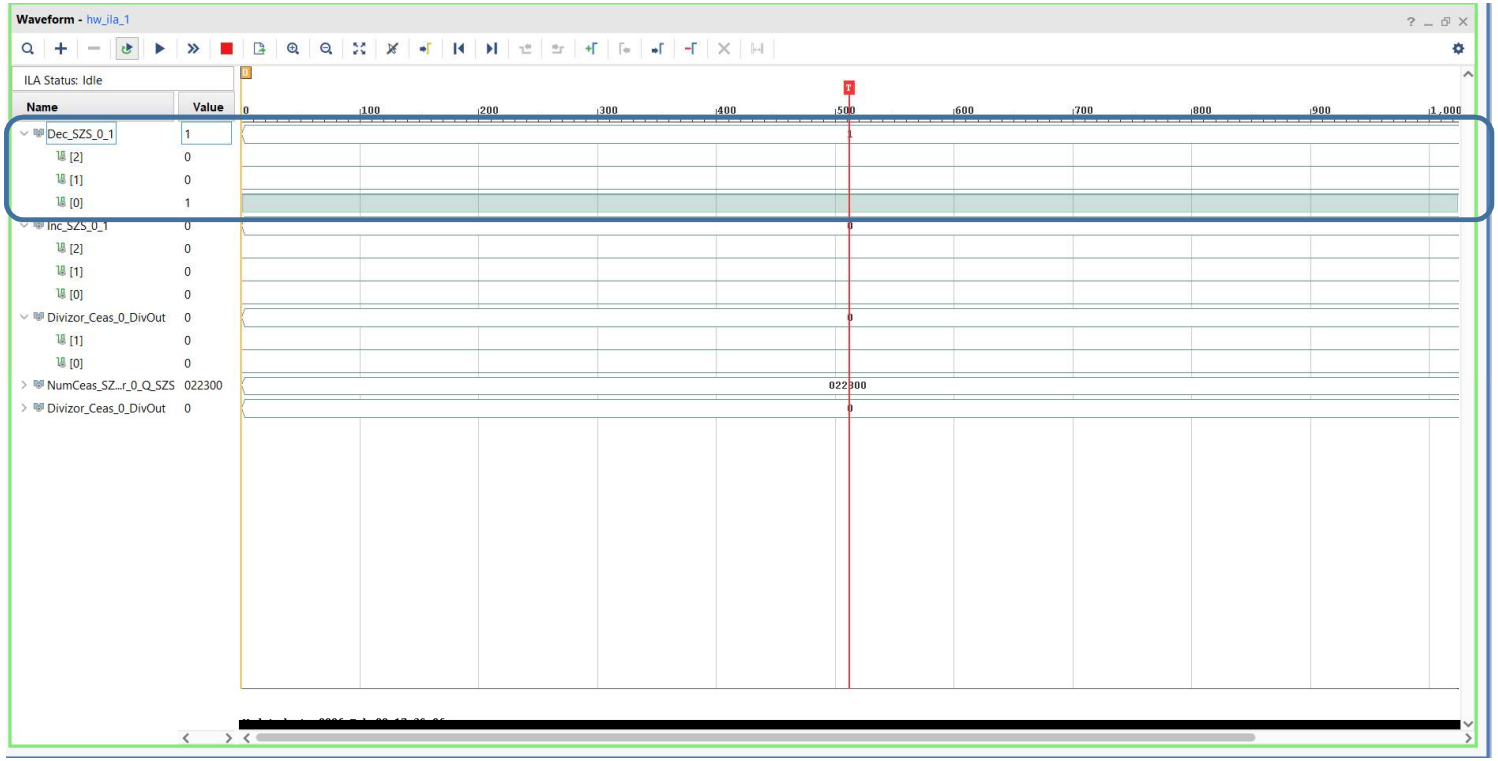
Adjustment of hours (from BTNR and BTNL)



Increment buttons – ILA screenshot



Decrement buttons – ILA screenshot



Demonstration that, after inserting the debouncers, the hours and minutes can be incremented or decremented

